5.5 Rational Angles That Aren't

WORKSHEET

The question

STEP III 2022, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Suppose θ were rational, say $\theta = \frac{p}{q}$ degrees. Find a positive integer k with $\sin(k\theta) = 0$.

STAGE 1 Multiple-angle formulas

Covers: Part (i)

Pause and try Write $\cos k\theta + \mathrm{i}\sin k\theta = (\cos\theta + \mathrm{i}\sin\theta)^k$ and take out a factor of $\cos^k\theta$. Expand $(1 + \mathrm{i}\tan\theta)^k$ and take imaginary and real parts. Then use $\tan^2\theta = \sec^2\theta - 1$.

STAGE 2 Parity

Covers: Part (ii)

Ask yourself With $\sec\theta = a$, an odd integer, is $\sec^2\theta - 1$ odd or even?

Pause and try Take the smallest k with $\sin(k\theta) = 0$. Then the bracket in the sine formula must be zero. Why does that force k to be even? Then use $\sin k\theta = 2\sin\frac{k\theta}{2}\cos\frac{k\theta}{2}$.

STAGE 3 The contradiction

Covers: Part (ii)

Pause and try Apply the cosine formula with $m = \frac{k}{2}$. Why can the bracket never be zero?

STAGE 4 The cotangent version

Covers: Part (iii)

Ask yourself If $\cot\phi = \frac{1}{b}$, what is $\sec^2\phi - 1$? Is it even? Then run the same argument.

What's really going on

Irrational rotations

If θ is a rational number of degrees, repeatedly rotating by θ eventually returns you to where you started. If θ is irrational, you never return, and something remarkable happens: the points $k\theta$ (taken modulo 360°) spread out *evenly* around the circle, coming arbitrarily close to every angle. This is the *equidistribution theorem*, proved independently by Hermann Weyl, Wacław Sierpiński and Piers Bohl around 1910. Figure 5.5.1 shows the first 40 multiples of $\cos^{-1}\frac{1}{3}$: they are already filling the circle, and none ever repeats.

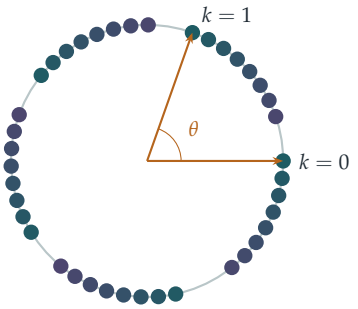


Figure 5.5.1 The points at angles $k\theta$, for $k = 0, 1, \ldots, 39$, where $\theta = \cos^{-1}\frac{1}{3} \approx 70.53^\circ$ (shading from teal to plum as k increases). Because θ is irrational, no point is ever repeated, and the points spread evenly around the whole circle.

Hilbert's third problem

In 1900 David Hilbert listed twenty-three problems for the new century. The third asked: if two polyhedra have the same volume, can one always be cut into finitely many polyhedral pieces and reassembled into the other? (For polygons in the plane the answer is yes: this is the Wallace–Bolyai–Gerwien theorem.) Within a year, Hilbert's student Max Dehn showed the answer is *no*: a regular tetrahedron cannot be cut up and reassembled into a cube of the same volume. Dehn's proof attaches to each polyhedron a quantity built from its edge lengths and dihedral angles, now called the *Dehn invariant*, which is unchanged by cutting and reassembling; for a cube it is zero, and for a regular tetrahedron it is non-zero precisely because the dihedral angle $\cos^{-1}\frac{1}{3}$ is not a rational multiple of 180°. That is exactly what this question proves.

Niven's theorem

How far does this go? In 1956 Ivan Niven proved that if θ is a rational number of degrees and $\cos\theta$ is rational, then $\cos\theta$ is one of $0, \pm\frac{1}{2}, \pm 1$. So the only "nice" angles with rational cosines are 0°, 60°, 90°, 120°, 180° and their relatives. The proof uses the fact that $2\cos\theta$ is a root of a monic polynomial with integer coefficients, an *algebraic integer*, the idea mentioned at the end of Section 1.3.

Try this next

1. **The tetrahedron.** Show that the angle between two faces of a regular tetrahedron is $\cos^{-1}\frac{1}{3}$, and that the angle between two bonds in a methane molecule (from the centre of a regular tetrahedron to two vertices) is $\cos^{-1}\left(-\frac{1}{3}\right)$.

2. **The 3–4–5 triangle.** Show that the smallest angle of a 3–4–5 triangle is an irrational number of degrees. (*Hint:* it is $\cot^{-1}\frac{4}{3}$; adapt the argument, or consider powers of the Gaussian integer $4+3\mathrm{i}$ and show none is real.)

3. **Why odd a?** What goes wrong in the argument when $a = 2$? Is $\cos^{-1}\frac{1}{2}$ rational?

4. **Niven for small cases.** Find all whole numbers of degrees θ between 0° and 90° for which $\cos\theta$ is rational, and check that your list agrees with Niven's theorem.

Further reading

- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). Includes Niven's theorem on rational values of trigonometric functions.
- **Martin Aigner & Günter Ziegler, *Proofs from THE BOOK***, chapter on Hilbert's third problem. A beautiful account of Dehn's solution.
- **Benjamin Yandell, *The Honors Class: Hilbert's Problems and Their Solvers*** (A. K. Peters). The stories behind all twenty-three problems.

The full guided solution is in Section 5.5:
https://beyondstep.keithmcnulty.org/ch05/rational-angles.html